NEXUS Z

Graph-Based Fraud Ring Detection on IBM LinuxONE

Solution Document | IBM Z Datathon 2026 | Theme: Real-Time AI for Critical Decisions

1. Problem

Pig-butchering scams groom victims over weeks and then push them to deposit money on fake investment platforms, often wiping out their savings. Banks score payments one at a time, and each deposit looks ordinary on its own, so the fraud passes undetected. The real signal is in the network: many victims paying a few collector accounts, funds layered through chains of mule accounts, and rapid pass-through to cash-out or crypto off-ramps. Because victims send money voluntarily, rules built around stolen credentials or unusual devices also fail to trigger. Investigators then trace links manually after the money has moved, and alerts without reasons are hard to act on.

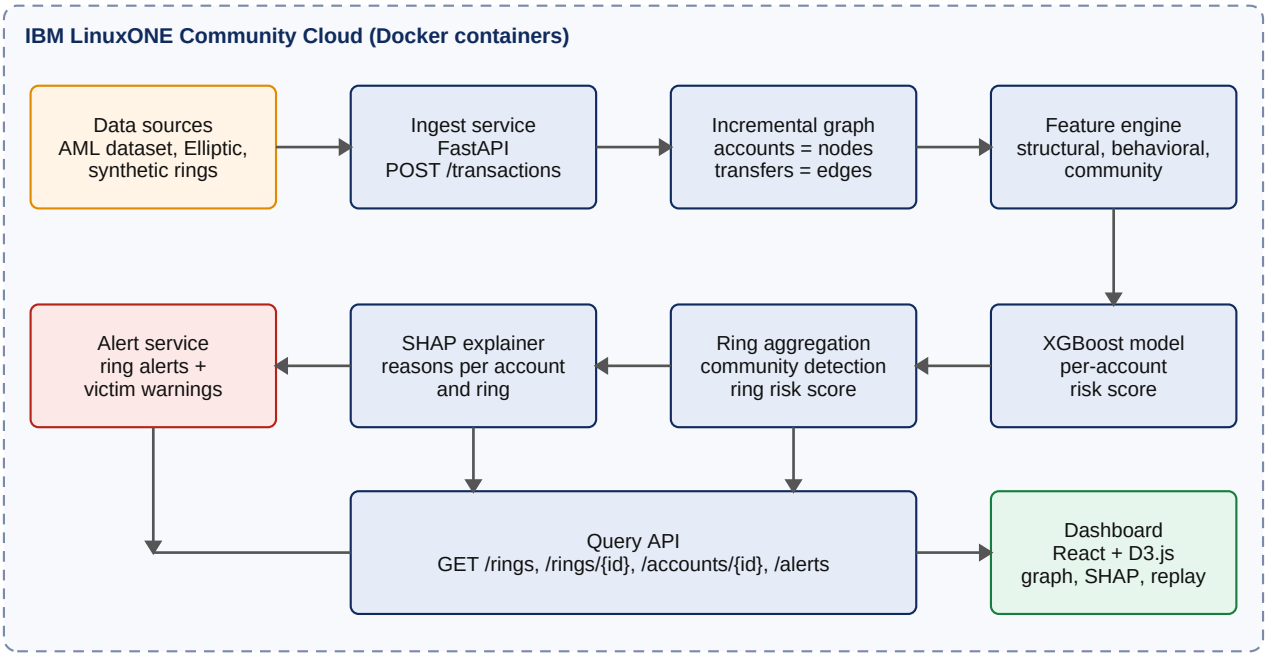
2. Objective and Scope

- Detect coordinated fraud rings in streaming transaction data, not isolated payments.
- Explain every flagged account and ring in terms an investigator can act on.
- Warn likely victims before further money is sent.
- Run end to end on IBM LinuxONE Community Cloud, using public and synthetic, non-personal data only.
- Scope for the 24-hour datathon: a working prototype with one scam typology (pig-butchering), one model family (XGBoost), and a live dashboard.

3. Solution Overview

NEXUS Z treats the transaction network as a time-evolving graph. Accounts are nodes; time-stamped, amount-weighted transfers are edges; shared device, IP or beneficiary links can connect related accounts. For each account the system computes graph and behavioral features, scores it with an XGBoost model, groups accounts into communities, and aggregates scores into a ring-level risk. SHAP values explain each score. When a ring is flagged, accounts that show the early fan-out pattern toward it are marked as likely victims and a pre-payment warning is raised.

4. Architecture



Transactions enter through the ingest service, update the in-memory graph incrementally, and trigger feature recomputation only for affected neighborhoods. The model scores those accounts, ring aggregation updates the community scores, and the explainer attaches reasons. The query API serves the dashboard and the alert service.

5. Data

Source	Role	Notes
Public synthetic AML transaction dataset (e.g., IBM AML transactions on Kaggle)	Realistic background traffic and baseline laundering patterns	Confirm availability and license before use
Public labeled cryptocurrency transaction graph (e.g., Elliptic Bitcoin dataset)	Test illicit fund-flow patterns on a real graph	Confirm availability and license before use
Synthetic pig-butchering rings generated by the team	Ground truth for ring recovery and victim-side patterns	Victim fan-in to collectors, mule chains, cash-out off-ramps; vary sizes, timing and shapes
Hard negatives generated by the team	Test false positives	Merchants, payroll, landlords, charities that look like hubs or rings

All identifiers are pseudonymized. No real personal data is used. Train/test split is by time.

6. Detection Method

6.1 Features

Group	Features	What they capture
Structural	In/out degree, fan-in/fan-out ratio, PageRank, k-core, clustering coefficient, 2-3 hop neighborhood statistics	Collector hubs, mule chains, dense coordinated groups
Behavioral	Transaction velocity, burst timing, pass-through time (money in, then out), amount splitting, account age	Layering and rapid cash-out
Community	Community id (Louvain/Leiden), community size and density, share of high-risk members	Ring-level context for each account

6.2 Scoring

- **Account score:** XGBoost classifier on the features above, trained with class weighting for the heavy imbalance.
- **Ring score:** aggregate of member scores, community density and flow concentration into a single ring risk.
- **Victim identification:** accounts whose recent outflows converge on a flagged ring and match the early fan-out pattern.

6.3 Explainability

SHAP values provide a per-account breakdown (for example, high fan-in, short pass-through time, new account age) and a ring-level summary of the top contributing signals.

7. Real-Time Serving

Endpoint	Purpose
POST /transactions	Ingest a transaction or batch; update the graph incrementally
GET /accounts/{id}	Risk score, SHAP reasons, neighborhood summary
GET /rings	List of detected rings with ring risk and size
GET /rings/{id}	Ring members, edges, fund-flow timeline, top reasons
GET /alerts	Ring alerts and victim warnings, newest first

8. Dashboard

- **Graph view:** force-directed D3.js graph; node color by risk, node size by flow volume; rings outlined.
- **Alert queue:** ranked rings and victim warnings with a one-line reason each.
- **Detail panel:** SHAP contribution chart, key metrics, member list for the selected account or ring.
- **Time slider:** replay fund flow to show how the ring formed and where money exited.
- **Victim warning preview:** the message a customer would see before a flagged payment.

9. Privacy and Security

- Graph and dashboard use pseudonymized account tokens; no names, contact details or raw personal data.
- Data stays inside the LinuxONE environment used for scoring.
- Future extension: institutions share ring signatures (patterns and hashed indicators) instead of raw records.

10. Evaluation Plan

Level	Metric	Purpose
Account	Precision, recall, F1, PR-AUC	Accuracy under heavy class imbalance
Ring	Ring recovery rate, purity, time-to-detection	Whether whole rings are found, and how early
System	End-to-end scoring latency on LinuxONE	Real-time claim, measured
Ablation	Transaction-only features vs. graph features	Shows what the graph adds
Robustness	Hard-negative false-positive rate; varied ring shapes	Guards against circular, overfit results

Results will be reported only as measured. Time-based splits prevent leakage, and the synthetic ring generator uses varied shapes so the model is not graded only on patterns it was built to find.

11. 24-Hour Build Plan

Window	Deliverable
Hours 0-3	L1CC setup, repo, dependency check on s390x, data loading, synthetic ring generator
Hours 3-8	Graph construction, features, baseline (transaction-only) model
Hours 8-13	Graph-feature model, ring aggregation, SHAP, evaluation numbers
Hours 13-18	FastAPI service, streaming simulator, incremental updates, latency measurement
Hours 18-22	React + D3.js dashboard, victim warning preview, demo scenario
Hours 22-24	Pitch deck, README, final testing, submission on Hack2Skill

12. Risks and Mitigations

Risk	Mitigation
Package incompatibility on s390x (big-endian)	Test XGBoost, SHAP and numpy builds on L1CC before the event; pin working versions
Circular evaluation on self-generated rings	Hard negatives, varied ring shapes, public datasets as background, time-based split
Label leakage through community features	Compute features using only edges available at scoring time
Scale limits on Community Cloud	Fix graph size early; report the size and latency actually measured

Risk	Mitigation
Scope creep in 24 hours	Single typology, single model family; GNN and consortium mode stay as future scope

13. Expected Outcome and Future Scope

- **Outcome:** a working prototype that detects injected fraud rings in streaming data, explains each flag, and raises victim warnings, running on IBM LinuxONE.
- **Future scope:** consortium ring-signature sharing across institutions; larger graphs; a GNN (GraphSAGE) compared against the feature-based model; additional scam typologies.